

3(a)	p = pressure (of gas), V = volume (of gas) and k = Boltzmann constant	B1
	N = number of molecules	B1
	T = thermodynamic temperature	B1
3(b)	$(pV = NkT \text{ and } pV = \frac{1}{3}Nm\langle c^2 \rangle \text{ leading to } NkT = \frac{1}{3}Nm\langle c^2 \rangle)$	M1
	algebra leading to $(3/2)kT = \frac{1}{2}m\langle c^2 \rangle$ and use of $\frac{1}{2}m\langle c^2 \rangle = E_K$ leading to $(3/2)kT = E_K$	A1
3(c)(i)	$T = 296 \text{ K}$	C1
	$\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$	C1
	$\frac{1}{2} \times 5.31 \times 10^{-26} \times u^2 = (3/2) \times 1.38 \times 10^{-23} \times 296$	
	$u = 480 \text{ m s}^{-1}$	A1
3(c)(ii)	line passing through (P, u)	B1
	horizontal straight line	B1